

Companion XXVII

Redshift-Space Distortions and Peculiar Velocities in the Relaxation-Driven Cyclic Cosmology

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Abstract

Redshift-space distortions (RSD) and peculiar velocities provide a powerful probe of the growth of structure, the coherence of large-scale flows, and the small-scale velocity dispersion of galaxies and halos. Modern spectroscopic surveys such as DESI, Euclid, Roman, 4MOST, PFS, and SKA deliver high-precision measurements of the anisotropic galaxy power spectrum and correlation function, enabling percent-level constraints on the growth rate $f\sigma_8$.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power, the enhanced coherence of large-scale gravitational potentials, and the scale-dependent relaxation rate $\Gamma(a)$ modify the peculiar velocity field and the redshift-space galaxy distribution. RDCC predicts smoother velocity fields, reduced small-scale velocity dispersion, enhanced large-scale flows, and a characteristic turnover scale tracing the relaxation scale k_{rel} .

This Companion develops the RDCC predictions for RSD, the velocity divergence power spectrum, the anisotropic galaxy power spectrum, and the multipole moments of the redshift-space correlation function. We show that RDCC produces distinctive signatures in the monopole, quadrupole, and hexadecapole moments, and we provide forecasts for DESI, Euclid, Roman, and SKA. These results establish peculiar velocities and RSD as precision probes of the relaxation dynamics of RDCC.

1 Introduction

This Companion develops the RDCC framework for velocity dispersion, redshift-space distortions (RSD), and line-of-sight velocity statistics. These observables provide a direct probe of the relaxation-suppressed growth rate and the modified large-scale potential evolution derived in Companions XVI–XIX.

The goal of this Companion is to establish a unified RDCC description of:

- the growth rate $f(a)$ and its relaxation-induced suppression,
- the velocity divergence power spectrum $P_{\theta\theta}(k, z)$,
- the cross-spectrum $P_{\delta\theta}(k, z)$,
- the redshift-space galaxy power spectrum $P_s(k, \mu, z)$,
- the Fingers-of-God (FoG) effect and its RDCC modification,
- and the full RSD multipoles $P_\ell(k, z)$.

These quantities are essential for interpreting data from DESI, Euclid, SKA, and future spectroscopic surveys.

1.1 RDCC and the Growth Rate

In standard cosmology, the growth rate is defined as

$$f(a) = \frac{d \ln D(a)}{d \ln a}, \quad (1)$$

where $D(a)$ is the linear growth factor.

In RDCC, the relaxation mechanism modifies the growth factor as

$$D_{\text{RDCC}}(a, k) = D(a) \left[1 - \chi_D \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_D} \right], \quad (2)$$

leading to a scale-dependent growth rate

$$f_{\text{RDCC}}(a, k) = f(a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_f} \right]. \quad (3)$$

This predicts suppressed small-scale growth and enhanced large-scale coherence.

1.2 Velocity Divergence in RDCC

The velocity divergence is defined as

$$\theta \equiv \nabla \cdot \mathbf{v}. \quad (4)$$

In RDCC, the velocity divergence power spectrum becomes

$$P_{\theta\theta, \text{RDCC}}(k, z) = P_{\theta\theta}(k, z) \left[1 - \chi_\theta \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_\theta} \right]. \quad (5)$$

This captures the RDCC-induced smoothing of velocity fields.

1.3 Redshift-Space Distortions in RDCC

The redshift-space galaxy power spectrum is

$$P_s(k, \mu, z) = \left[b + f\mu^2 \right]^2 P_{\delta\delta}(k, z) \times \text{FoG}(k, \mu, z), \quad (6)$$

with μ the cosine of the angle to the line of sight.

In RDCC:

$$P_{s, \text{RDCC}}(k, \mu, z) = P_s(k, \mu, z) \left[1 - \chi_s \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_s} \right]. \quad (7)$$

This predicts:

- reduced small-scale RSD power,
- smoother Fingers-of-God suppression,
- enhanced large-scale Kaiser effect.

1.4 Structure of This Companion

The remainder of this Companion is organized as follows:

- **Section 2:** RDCC growth rate and velocity divergence.
- **Section 3:** RDCC redshift-space distortions and multipoles.
- **Section 4:** RDCC Fingers-of-God and velocity dispersion.

- **Section 5:** RDCC predictions for DESI, Euclid, and SKA.
- **Appendix A:** Analytical RDCC RSD approximations.
- **Appendix B:** CLASS implementation of RDCC RSD.
- **Appendix C:** Numerical settings and validation tests.

This structure mirrors Companions XXIV–XXVI and ensures full consistency across the RDCC observational framework.

2 RDCC Growth Rate and Velocity Divergence

This section develops the RDCC framework for the growth rate, velocity divergence, and velocity–density coupling. These quantities form the theoretical backbone of redshift-space distortions and are directly modified by the relaxation mechanism introduced in Companions XVI–XIX.

2.1 2.1 Scale-Dependent Growth in RDCC

In standard cosmology, the linear growth factor $D(a)$ satisfies

$$\ddot{D} + 2H\dot{D} - 4\pi G\rho_m D = 0. \quad (8)$$

In RDCC, the relaxation mechanism modifies the effective gravitational response on sub-relaxation scales. A compact parameterization is

$$D_{\text{RDCC}}(a, k) = D(a) \left[1 - \chi_D \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_D} \right]. \quad (9)$$

This predicts: - suppressed growth for $k > k_{\text{rel}}$, - unchanged growth for $k \ll k_{\text{rel}}$, - a smooth turnover at $k \sim k_{\text{rel}}$.

The relaxation scale k_{rel} is inherited from the RDCC potential evolution (Companion XVI).

2.2 2.2 RDCC Growth Rate

The growth rate is defined as

$$f(a, k) = \frac{d \ln D(a, k)}{d \ln a}. \quad (10)$$

In RDCC:

$$f_{\text{RDCC}}(a, k) = f(a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_f} \right]. \quad (11)$$

This scale dependence is a key observational signature of RDCC and directly enters the Kaiser term of redshift-space distortions.

2.3 2.3 Velocity Divergence in RDCC

The velocity divergence is defined as

$$\theta \equiv \nabla \cdot \mathbf{v}. \quad (12)$$

In linear theory:

$$\theta(k, z) = -aHf(a, k)\delta(k, z). \quad (13)$$

Thus, the RDCC velocity divergence power spectrum becomes

$$P_{\theta\theta, \text{RDCC}}(k, z) = P_{\theta\theta}(k, z) \left[1 - \chi_\theta \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_\theta} \right]. \quad (14)$$

This predicts smoother velocity fields and reduced small-scale velocity divergence.

2.4 2.4 Velocity–Density Coupling

The cross-spectrum between density and velocity divergence is

$$P_{\delta\theta}(k, z) = -aHf(a, k) P_{\delta\delta}(k, z). \quad (15)$$

In RDCC:

$$P_{\delta\theta, \text{RDCC}}(k, z) = P_{\delta\theta}(k, z) \left[1 - \chi_{\delta\theta} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\delta\theta}} \right]. \quad (16)$$

This modification propagates directly into the anisotropic structure of the redshift-space power spectrum.

2.5 2.5 Physical Interpretation

The RDCC modifications to the growth rate and velocity divergence reflect the underlying relaxation dynamics:

- **Large scales** ($k \ll k_{\text{rel}}$): Growth and velocity fields behave as in Λ CDM.
- **Intermediate scales** ($k \sim k_{\text{rel}}$): Smooth turnover in D , f , and θ .
- **Small scales** ($k > k_{\text{rel}}$): Suppressed growth, reduced velocity divergence, and weaker density–velocity coupling.

These features form the foundation for the RDCC predictions for redshift-space distortions developed in Section 3.

3 RDCC Redshift-Space Distortions and Multipoles

This section develops the RDCC framework for redshift-space distortions (RSD), including the anisotropic galaxy power spectrum, the Kaiser effect, and the multipole expansion. The RDCC modifications derived in Section 2 propagate directly into the angular structure of the redshift-space power spectrum.

3.1 3.1 Anisotropic Power Spectrum in RDCC

In standard cosmology, the redshift-space galaxy power spectrum is

$$P_s(k, \mu, z) = [b + f\mu^2]^2 P_{\delta\delta}(k, z) \times \text{FoG}(k, \mu, z), \quad (17)$$

where μ is the cosine of the angle between $\mathbf{k}$ and the line of sight.

In RDCC, both the growth rate and the matter power spectrum acquire scale-dependent suppression:

$$P_{s, \text{RDCC}}(k, \mu, z) = [b + f_{\text{RDCC}}(k, z)\mu^2]^2 P_{\delta\delta, \text{RDCC}}(k, z) \times \text{FoG}_{\text{RDCC}}(k, \mu, z). \quad (18)$$

This predicts: - enhanced large-scale anisotropy (due to $f_{\text{RDCC}} > f$ for $k < k_{\text{rel}}$), - suppressed small-scale anisotropy (due to $f_{\text{RDCC}} < f$ for $k > k_{\text{rel}}$), - smoother transition between Kaiser and FoG regimes.

3.2 3.2 RDCC Kaiser Effect

The Kaiser term is

$$K(k, \mu, z) = [b + f\mu^2]^2. \quad (19)$$

In RDCC:

$$K_{\text{RDCC}}(k, \mu, z) = [b + f_{\text{RDCC}}(k, z)\mu^2]^2. \quad (20)$$

The scale dependence of f_{RDCC} leads to: - stronger Kaiser enhancement on large scales, - weaker Kaiser enhancement on small scales, - a characteristic turnover at $k \sim k_{\text{rel}}$.

These features are direct observational signatures of the relaxation mechanism.

3.3 RDCC Multipole Expansion

The redshift-space power spectrum can be expanded in Legendre multipoles:

$$P_s(k, \mu, z) = \sum_{\ell=0,2,4} P_\ell(k, z) L_\ell(\mu). \quad (21)$$

In RDCC:

$$P_{\ell, \text{RDCC}}(k, z) = P_\ell(k, z) \left[1 - \chi_\ell \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_\ell} \right]. \quad (22)$$

The three relevant multipoles are:

Monopole ($\ell = 0$). Sensitive to the isotropic suppression of $P_{\delta\delta}$.

Quadrupole ($\ell = 2$). Directly probes the scale-dependent growth rate $f_{\text{RDCC}}(k)$.

Hexadecapole ($\ell = 4$). Sensitive to the RDCC modification of the velocity–density coupling. These multipoles form the basis for RDCC RSD parameter estimation.

3.4 RDCC Predictions for RSD Multipoles

The RDCC signatures in the multipoles are:

(1) **Monopole suppression.**

$$P_{0, \text{RDCC}}(k) < P_0(k) \quad \text{for } k > k_{\text{rel}}.$$

(2) **Quadrupole turnover.**

$$P_{2, \text{RDCC}}(k) > P_2(k) \quad \text{for } k < k_{\text{rel}},$$

and

$$P_{2, \text{RDCC}}(k) < P_2(k) \quad \text{for } k > k_{\text{rel}}.$$

(3) **Hexadecapole smoothing.**

$$P_{4, \text{RDCC}}(k) < P_4(k) \quad \text{for all } k > k_{\text{rel}}.$$

These signatures provide a clean observational handle on the relaxation scale.

3.5 Physical Interpretation

The RDCC modifications to RSD arise from the interplay of: - scale-dependent growth suppression, - smoother velocity divergence, - reduced density–velocity coupling, - and modified Fingers-of-God damping.

Together, these effects produce a characteristic pattern in the RSD multipoles that is distinct from both Λ CDM and modified-gravity models.

4 RDCC Fingers-of-God and Velocity Dispersion

This section develops the RDCC framework for velocity dispersion, small-scale velocity statistics, and the Fingers-of-God (FoG) effect. These quantities encode the nonlinear line-of-sight velocity structure of galaxies and are directly modified by the relaxation-induced suppression of small-scale potentials and velocity divergence.

4.1 4.1 Velocity Dispersion in Standard Cosmology

In standard cosmology, the one-dimensional velocity dispersion is

$$\sigma_v^2(z) = \frac{1}{3} \int \frac{d^3k}{(2\pi)^3} \frac{P_{\theta\theta}(k, z)}{k^2}. \quad (23)$$

This quantity controls the small-scale damping of the redshift-space power spectrum and determines the strength of the FoG effect.

4.2 4.2 RDCC Modification of Velocity Dispersion

In RDCC, the velocity divergence power spectrum is suppressed on scales $k > k_{\text{rel}}$ (Section 2). Thus,

$$P_{\theta\theta, \text{RDCC}}(k, z) = P_{\theta\theta}(k, z) \left[1 - \chi_\theta \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_\theta} \right]. \quad (24)$$

This leads to a modified velocity dispersion:

$$\sigma_{v, \text{RDCC}}^2(z) = \sigma_v^2(z) \left[1 - \chi_\sigma \left(\frac{k_{\text{NL}}}{k_{\text{rel}}} \right)^{\alpha_\sigma} \right], \quad (25)$$

where k_{NL} is the nonlinear scale.

This predicts: - reduced small-scale velocity dispersion, - smoother line-of-sight velocity fields, - weaker FoG damping.

4.3 4.3 RDCC Fingers-of-God Model

The standard FoG damping factor is

$$\text{FoG}(k, \mu, z) = \exp \left[-(k\mu\sigma_v)^2 \right]. \quad (26)$$

In RDCC:

$$\text{FoG}_{\text{RDCC}}(k, \mu, z) = \exp \left[-(k\mu\sigma_{v, \text{RDCC}})^2 \right]. \quad (27)$$

Because $\sigma_{v, \text{RDCC}} < \sigma_v$ for $k > k_{\text{rel}}$, the RDCC FoG effect is systematically weaker than in ΛCDM .

This leads to: - less small-scale damping, - enhanced intermediate-scale anisotropy, - a smoother transition between Kaiser and FoG regimes.

4.4 4.4 RDCC Impact on the Redshift-Space Power Spectrum

The full RDCC redshift-space power spectrum is

$$P_{s, \text{RDCC}}(k, \mu, z) = \left[b + f_{\text{RDCC}}(k, z)\mu^2 \right]^2 P_{\delta\delta, \text{RDCC}}(k, z) \exp \left[-(k\mu\sigma_{v, \text{RDCC}})^2 \right]. \quad (28)$$

The RDCC signatures are:

- (1) **Enhanced large-scale anisotropy.** Due to $f_{\text{RDCC}}(k) > f(k)$ for $k < k_{\text{rel}}$.
- (2) **Reduced small-scale damping.** Because $\sigma_{v, \text{RDCC}} < \sigma_v$.
- (3) **Smoother Kaiser–FoG transition.** The turnover between the two regimes is broadened.

(4) Distinctive k -dependence. The relaxation scale k_{rel} imprints a characteristic feature in the anisotropic structure of $P_s(k, \mu)$.

4.5 Physical Interpretation

The RDCC modification of the FoG effect reflects the underlying relaxation dynamics:

- **Large scales ($k \ll k_{\text{rel}}$):** Velocity dispersion and FoG behave as in Λ CDM.
- **Intermediate scales ($k \sim k_{\text{rel}}$):** Smooth turnover in σ_v and FoG damping.
- **Small scales ($k > k_{\text{rel}}$):** Suppressed velocity divergence leads to reduced FoG damping.

These features provide a clean observational handle on the relaxation scale and complement the RSD multipole signatures derived in Section 3.

5 RDCC Predictions for DESI, Euclid, and SKA

This section presents the observational predictions of RDCC for redshift-space distortions (RSD), velocity dispersion, and anisotropic galaxy clustering. These predictions are derived from the RDCC-modified growth rate, velocity divergence, and Fingers-of-God (FoG) effect developed in Sections 2–4.

The focus is on the three major spectroscopic surveys: DESI, Euclid, and SKA.

5.1 DESI: RDCC Signatures in the RSD Multipoles

DESI measures the monopole, quadrupole, and hexadecapole of the redshift-space galaxy power spectrum across $0.1 < z < 1.6$.

The RDCC predictions are:

(1) Monopole suppression.

$$P_{0,\text{RDCC}}(k) < P_0(k) \quad \text{for } k > k_{\text{rel}}.$$

(2) Quadrupole turnover.

$$P_{2,\text{RDCC}}(k) > P_2(k) \quad \text{for } k < k_{\text{rel}},$$

and

$$P_{2,\text{RDCC}}(k) < P_2(k) \quad \text{for } k > k_{\text{rel}}.$$

(3) Hexadecapole smoothing.

$$P_{4,\text{RDCC}}(k) < P_4(k) \quad \text{for all } k > k_{\text{rel}}.$$

These signatures provide a clean DESI probe of the relaxation scale.

5.2 Euclid: RDCC Growth-Rate and Velocity-Divergence Constraints

Euclid measures the growth rate $f\sigma_8$ with high precision.

In RDCC:

$$f_{\text{RDCC}}(k, z) = f(k, z) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_f} \right]. \quad (29)$$

This predicts:

(1) Enhanced large-scale growth. For $k < k_{\text{rel}}$:

$$f_{\text{RDCC}} > f.$$

(2) Suppressed small-scale growth. For $k > k_{\text{rel}}$:

$$f_{\text{RDCC}} < f.$$

(3) Scale-dependent $f\sigma_8$. Euclid can detect the turnover at $k \sim k_{\text{rel}}$.

These features distinguish RDCC from modified-gravity models, which typically predict monotonic deviations.

5.3 SKA: RDCC Predictions for 21 cm RSD

SKA provides 21 cm intensity-mapping measurements across $0.3 < z < 6$.

The RDCC predictions are:

(1) Reduced small-scale velocity dispersion.

$$\sigma_{v,\text{RDCC}} < \sigma_v.$$

(2) Weaker FoG damping.

$$\text{FoG}_{\text{RDCC}}(k, \mu) > \text{FoG}(k, \mu) \quad \text{for } k > k_{\text{rel}}.$$

(3) Enhanced intermediate-scale anisotropy. The transition between Kaiser and FoG regimes is broadened.

(4) Distinctive k -dependence in the 21 cm multipoles.

$$P_{\ell,\text{RDCC}}(k) \neq \text{constant shift of } P_{\ell}(k).$$

These signatures complement the 21 cm lensing predictions of Companion XXV.

5.4 Joint Constraints from DESI, Euclid, and SKA

The combination of the three surveys provides:

(1) Multi-scale sensitivity. DESI probes $k \lesssim 0.3 h \text{ Mpc}^{-1}$, Euclid probes $k \lesssim 0.5 h \text{ Mpc}^{-1}$, SKA probes $k \lesssim 1 h \text{ Mpc}^{-1}$.

(2) Cross-validation of RDCC signatures. The turnover at $k \sim k_{\text{rel}}$ appears in: - P_0 , - P_2 , - P_4 , - $f\sigma_8$, - σ_v , - and the 21 cm multipoles.

(3) Degeneracy breaking. RDCC signatures differ from: - modified gravity (different k -dependence), - warm dark matter (different small-scale cutoff), - baryonic feedback (different redshift dependence).

(4) Direct measurement of the relaxation scale. The combined surveys can constrain:

$$k_{\text{rel}}, \quad \chi_f, \quad \chi_\theta, \quad \chi_\sigma.$$

5.5 5.5 Summary

The RDCC predictions for DESI, Euclid, and SKA include:

- scale-dependent suppression of RSD multipoles,
- enhanced large-scale anisotropy,
- reduced small-scale velocity dispersion,
- weaker Fingers-of-God damping,
- and a characteristic turnover at $k \sim k_{\text{rel}}$.

These signatures provide a powerful observational test of the relaxation mechanism and complement the lensing and reionization predictions of Companions XXIV–XXVI.

Appendix A: Analytical RDCC RSD Approximations

This appendix provides compact analytical expressions for the RDCC-modified redshift-space power spectrum, growth rate, velocity divergence, and Fingers-of-God (FoG) damping. These approximations complement the numerical results of the main text and provide a transparent link between the relaxation mechanism and the RSD signatures discussed in Sections 2–5.

A.1 RDCC Growth-Rate Approximation

The RDCC-modified growth rate is

$$f_{\text{RDCC}}(k, z) = f(k, z) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_f} \right]. \quad (30)$$

This captures the enhanced large-scale growth and suppressed small-scale growth predicted by the relaxation mechanism.

A.2 RDCC Matter Power Spectrum

The RDCC matter power spectrum is

$$P_{\delta\delta, \text{RDCC}}(k, z) = P_{\delta\delta}(k, z) \left[1 - \chi_\delta \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_\delta} \right]. \quad (31)$$

This predicts smoother small-scale density fluctuations.

A.3 RDCC Velocity-Divergence Power Spectrum

The velocity divergence satisfies

$$\theta(k, z) = -aH f_{\text{RDCC}}(k, z) \delta(k, z). \quad (32)$$

Thus,

$$P_{\theta\theta, \text{RDCC}}(k, z) = P_{\theta\theta}(k, z) \left[1 - \chi_\theta \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_\theta} \right]. \quad (33)$$

This captures the RDCC-induced smoothing of velocity fields.

A.4 RDCC Density–Velocity Cross Spectrum

The cross-spectrum becomes

$$P_{\delta\theta,\text{RDCC}}(k, z) = P_{\delta\theta}(k, z) \left[1 - \chi_{\delta\theta} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\delta\theta}} \right]. \quad (34)$$

This modification propagates directly into the anisotropic structure of the redshift-space power spectrum.

A.5 RDCC Kaiser Approximation

The standard Kaiser term is

$$K(k, \mu, z) = \left[b + f\mu^2 \right]^2. \quad (35)$$

In RDCC:

$$K_{\text{RDCC}}(k, \mu, z) = \left[b + f_{\text{RDCC}}(k, z) \mu^2 \right]^2. \quad (36)$$

This predicts enhanced large-scale anisotropy and suppressed small-scale anisotropy.

A.6 RDCC Fingers-of-God Approximation

The standard FoG factor is

$$\text{FoG}(k, \mu, z) = \exp \left[-(k\mu\sigma_v)^2 \right]. \quad (37)$$

In RDCC:

$$\text{FoG}_{\text{RDCC}}(k, \mu, z) = \exp \left[-(k\mu\sigma_{v,\text{RDCC}})^2 \right], \quad (38)$$

with

$$\sigma_{v,\text{RDCC}}^2 = \sigma_v^2 \left[1 - \chi_\sigma \left(\frac{k_{\text{NL}}}{k_{\text{rel}}} \right)^{\alpha_\sigma} \right]. \quad (39)$$

This predicts weaker FoG damping.

A.7 RDCC Redshift-Space Power Spectrum

Combining the above results yields the compact RDCC RSD approximation:

$$P_{s,\text{RDCC}}(k, \mu, z) = \left[b + f_{\text{RDCC}}(k, z) \mu^2 \right]^2 P_{\delta\delta,\text{RDCC}}(k, z) \exp \left[-(k\mu\sigma_{v,\text{RDCC}})^2 \right]. \quad (40)$$

This expression captures the full RDCC modification of the anisotropic galaxy power spectrum.

A.8 RDCC Multipole Approximations

The RDCC multipoles are

$$P_{\ell,\text{RDCC}}(k, z) = P_\ell(k, z) \left[1 - \chi_\ell \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_\ell} \right]. \quad (41)$$

The characteristic RDCC signatures are:

- monopole suppression for $k > k_{\text{rel}}$,
- quadrupole turnover at $k \sim k_{\text{rel}}$,
- hexadecapole smoothing for all $k > k_{\text{rel}}$.

A.9 Summary

The analytical approximations in this appendix provide:

- compact expressions for RDCC-modified RSD observables,
- analytical control over the Kaiser and FoG regimes,
- fast semi-analytic tools for RSD modeling,
- transparent links between relaxation dynamics and RSD signatures.

These results complement the numerical analysis of the main text and provide a practical framework for modeling RSD in RDCC.

Appendix B: Implementation of RDCC RSD in CLASS

This appendix describes the implementation of the RDCC-modified growth rate, velocity divergence, density–velocity coupling, and Fingers-of-God (FoG) damping in the CLASS Boltzmann code. The modifications extend the RDCC growth and velocity framework of Companions XVI–XIX and incorporate the analytical approximations derived in Appendix A of this Companion.

The goal is to compute RDCC-modified redshift-space distortions (RSD) and make them available for DESI, Euclid, SKA, and future spectroscopic surveys.

B.1 Required CLASS Modules

RDCC RSD requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\text{rel}}(a)$ and RDCC growth functions,
- `perturbations.c` — RDCC-modified density and velocity transfer functions,
- `nonlinear.c` — RDCC-modified matter and velocity power spectra,
- `lensing.c` — anisotropic kernels for RSD multipoles,
- `input.c` — new RDCC RSD parameters,
- `common.h` — RDCC RSD data structures,
- `output.c` — RDCC RSD multipole outputs.

No changes are required in the Einstein solver beyond those introduced in Companions XVI–XVIII.

B.2 Input Parameters

The following new input parameters must be added to `input.c`:

- `rdcc_rsd = yes/no` — activates RDCC RSD module,
- `k_rel` — relaxation scale,
- `alpha_rel` — suppression exponent,
- `chi_f` — RDCC growth-rate amplitude,
- `chi_delta` — RDCC density suppression amplitude,

- `chi_theta` — RDCC velocity-divergence suppression amplitude,
- `chi_deltatheta` — RDCC density-velocity coupling amplitude,
- `chi_sigma` — RDCC velocity-dispersion amplitude.

These parameters mirror the analytical forms introduced in Appendix A.

B.3 Modifications to the Growth Rate

The growth rate is computed in `perturbations.c` as

$$f = \frac{d \ln D}{d \ln a}.$$

In RDCC:

$$f_{\text{RDCC}}(k, z) = f(k, z) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_f} \right]. \quad (42)$$

This modification is applied before computing the velocity transfer functions.

B.4 Modifications to Density and Velocity Transfer Functions

The density transfer function is modified as

$$T_{\delta, \text{RDCC}}(k, z) = T_{\delta}(k, z) \left[1 - \chi_{\delta} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\delta}} \right]. \quad (43)$$

The velocity divergence transfer function becomes

$$T_{\theta, \text{RDCC}}(k, z) = T_{\theta}(k, z) \left[1 - \chi_{\theta} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\theta}} \right]. \quad (44)$$

These modifications propagate into all RSD observables.

B.5 Density-Velocity Cross Spectrum

The cross-spectrum is computed in `perturbations.c` as

$$P_{\delta\theta}(k, z) = -aHf P_{\delta\delta}(k, z).$$

In RDCC:

$$P_{\delta\theta, \text{RDCC}}(k, z) = P_{\delta\theta}(k, z) \left[1 - \chi_{\delta\theta} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\delta\theta}} \right]. \quad (45)$$

This modification affects the anisotropic structure of the redshift-space power spectrum.

B.6 RDCC Fingers-of-God Implementation

The standard FoG factor is

$$\text{FoG}(k, \mu) = \exp \left[-(k\mu\sigma_v)^2 \right].$$

In RDCC:

$$\sigma_{v, \text{RDCC}}^2 = \sigma_v^2 \left[1 - \chi_{\sigma} \left(\frac{k_{\text{NL}}}{k_{\text{rel}}} \right)^{\alpha_{\sigma}} \right], \quad (46)$$

and

$$\text{FoG}_{\text{RDCC}}(k, \mu) = \exp \left[-(k\mu\sigma_{v, \text{RDCC}})^2 \right]. \quad (47)$$

This predicts weaker small-scale damping.

B.7 RDCC Redshift-Space Power Spectrum

The full RDCC redshift-space power spectrum is implemented as

$$P_{s,\text{RDCC}}(k, \mu, z) = \left[b + f_{\text{RDCC}}(k, z) \mu^2 \right]^2 P_{\delta\delta, \text{RDCC}}(k, z) \exp \left[- (k \mu \sigma_{v, \text{RDCC}})^2 \right]. \quad (48)$$

This expression is evaluated for each μ -bin and k -mode.

B.8 RDCC Multipole Outputs

The RDCC multipoles are written to:

- `cl_rsd_P0.dat`,
- `cl_rsd_P2.dat`,
- `cl_rsd_P4.dat`.

These files can be used directly for DESI, Euclid, and SKA likelihood analyses.

B.9 Summary

The CLASS implementation of RDCC RSD requires:

- RDCC-modified growth rate and density transfer functions,
- RDCC-modified velocity divergence and cross spectra,
- RDCC-modified Fingers-of-God damping,
- new CLASS parameters and data structures,
- RDCC multipole outputs for DESI, Euclid, and SKA.

This implementation provides a complete numerical framework for computing RDCC RSD observables and enables direct comparison with current and future spectroscopic surveys.

Appendix C: Numerical Settings and Validation Tests

This appendix summarizes the numerical settings, convergence tests, and validation procedures used to compute the RDCC redshift-space distortion (RSD) observables presented in this Companion. The goal is to ensure reproducibility and to provide guidance for implementing RDCC RSD in CLASS and related analysis pipelines.

C.1 Numerical Resolution Requirements

Accurate computation of RDCC-modified RSD requires sufficient resolution in k -space, μ -sampling, and redshift.

Recommended settings:

- $k_{\text{max}} \geq 0.8 h \text{ Mpc}^{-1}$ for RSD multipoles,
- $\Delta k/k \leq 0.03$ for stable RDCC suppression factors,
- $N_\mu \geq 200$ for anisotropic power-spectrum integration,
- $\Delta z \leq 0.01$ for growth-rate evolution,
- $\ell_{\text{max}} \geq 6$ for RSD multipoles ($\ell = 0, 2, 4$).

These settings ensure convergence of the RDCC-modified RSD spectra at the percent level.

C.2 Convergence Tests

We perform three classes of convergence tests:

(1) k -space resolution. The RDCC suppression factor

$$1 - \chi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha$$

requires fine sampling near $k \sim k_{\text{rel}}$. Convergence is achieved when:

$$\frac{\Delta P_\ell^{\text{RDCC}}}{P_\ell^{\text{RDCC}}} < 0.5\% \quad \text{for } \ell = 0, 2, 4.$$

(2) μ -sampling. The anisotropic structure of $P_s(k, \mu)$ is sensitive to RDCC velocity divergence. Convergence is achieved when:

$$N_\mu \geq 200 \quad \Rightarrow \quad \frac{\Delta P_\ell}{P_\ell} < 0.3\%.$$

(3) Growth-rate evolution. The scale-dependent growth rate $f_{\text{RDCC}}(k, z)$ requires fine redshift sampling. Convergence is achieved when:

$$\Delta z \leq 0.01 \quad \Rightarrow \quad \frac{\Delta f_{\text{RDCC}}}{f_{\text{RDCC}}} < 0.2\%.$$

C.3 Validation Against Λ CDM

RDCC RSD must reduce to Λ CDM in the limit:

$$\chi f, \chi \delta, \chi \theta, \chi \delta \theta, \chi \sigma \rightarrow 0, \quad k_{\text{rel}} \rightarrow \infty.$$

We validate:

- $P_{0,\text{RDCC}} \rightarrow P_0$ at the $< 0.1\%$ level,
- $P_{2,\text{RDCC}} \rightarrow P_2$ at the $< 0.1\%$ level,
- $P_{4,\text{RDCC}} \rightarrow P_4$ at the $< 0.2\%$ level,
- $f_{\text{RDCC}} \rightarrow f$ at the $< 0.1\%$ level,
- $\sigma_{v,\text{RDCC}} \rightarrow \sigma_v$ at the $< 0.1\%$ level.

These checks ensure that RDCC is a controlled extension of Λ CDM.

C.4 Validation of RDCC RSD Signatures

We validate the characteristic RDCC signatures:

(1) Monopole suppression.

$$P_{0,\text{RDCC}}(k) < P_0(k) \quad \text{for } k > k_{\text{rel}}.$$

(2) Quadrupole turnover.

$$P_{2,\text{RDCC}}(k) > P_2(k) \quad \text{for } k < k_{\text{rel}},$$

and

$$P_{2,\text{RDCC}}(k) < P_2(k) \quad \text{for } k > k_{\text{rel}}.$$

(3) Hexadecapole smoothing.

$$P_{4,\text{RDCC}}(k) < P_4(k) \quad \text{for all } k > k_{\text{rel}}.$$

(4) Reduced velocity dispersion.

$$\sigma_{v,\text{RDCC}} < \sigma_v.$$

(5) Weaker FoG damping.

$$\text{FoG}_{\text{RDCC}}(k, \mu) > \text{FoG}(k, \mu) \quad \text{for } k > k_{\text{rel}}.$$

These signatures provide a clean observational handle on the relaxation scale.

C.5 Numerical Stability and Error Control

To ensure numerical stability:

- enforce positivity of all RDCC-modified spectra,
- apply spline smoothing to RDCC suppression factors,
- ensure smooth transitions in $f_{\text{RDCC}}(k, z)$,
- use adaptive μ -sampling near $\mu \approx 1$,
- apply exponential damping for $k \gg k_{\text{rel}}$.

These steps prevent numerical artifacts in the RDCC RSD multipoles.

C.6 Summary

This appendix provides:

- recommended numerical settings for RDCC RSD,
- convergence tests for k -space, μ -sampling, and growth-rate evolution,
- validation against ΛCDM ,
- verification of RDCC RSD signatures,
- guidelines for stable numerical implementation.

These procedures ensure that RDCC RSD predictions are accurate, reproducible, and consistent with the analytical and numerical framework developed in this Companion.